

Settled Weight: A Sealed, Zero-Parameter Test of a Predicted Weak-Field Acceleration Law on 175 SPARC Galaxies

Shaun Fosmark
Independent Researcher
sfosmark@proton.me

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Abstract

We test a predicted weak-field acceleration law on the 175 late-type galaxies of the SPARC database using a sealed, dual-implemented protocol with zero free parameters. The law predicts $g^2 = g_{\text{bar}}^2 + \lambda_{\text{eff}} a_S g_{\text{bar}}$ with a derived acceleration scale $a_S = cH_0/2\pi = 1.05 \times 10^{-10} \text{ m s}^{-2}$ and geometric coefficients bounded between $4/3$ (spherical sources) and $3\pi/8$ (disk sources), with no fitted quantities. Three results follow. (1) The data’s preferred deep-regime scale $\lambda^* a_S$ lies within 0.9% of the empirical radial-acceleration-relation scale a_0 . (2) A geometry-only model reaches 0.45% parity with the best globally fitted constant, wins four predeclared high-quality subgroups, and is falsified in gas-rich dwarfs under registered criteria. (3) A successor model that weights the anomalous boost by accumulated dynamical coherence—discounting turbulent gas by its ordered support fraction—defeats the best-fit constant on the full sample (0.097 vs. 0.099 dex in per-point $\log_{10} V$ RMS), helping 57% of galaxies individually, with quantified residuals in extreme-gas dwarfs reported as targets rather than fits. The central prediction—that the boost tracks accumulated dynamical coherence rather than instantaneous mass—implies weaker boosts in high-redshift disks, testable with existing kinematic surveys.

1 Introduction

Rotation curves of disk galaxies depart from Newtonian expectations below a characteristic acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, a regularity sharpened by the radial acceleration relation (RAR) on

the SPARC sample (McGaugh et al., 2016; Lelli et al., 2016). Milgrom noted at MOND’s founding that a_0 is numerically close to $cH_0/2\pi$ (Milgrom, 1983); the coincidence has remained underived in the phenomenology (Famaey & McGaugh, 2012), and emergent-gravity approaches place related scales at $cH_0/6$ (Verlinde, 2017).

Solvency Field Theory (SFT) is an independent research program in which spacetime geometry is the accumulated record of irreversible maintenance events sourced by matter (Fosmark, 2026). In its weak-field sector, SFT derives the acceleration scale $a_S = cH_0/2\pi$ from boundary thermodynamics and predicts that the coefficient λ multiplying the anomalous acceleration term is not universal but set by the source’s geometric structure, bounded by two parameter-free endpoints: $\lambda = 4/3$ for isotropic (spherical) sources and $\lambda = 3\pi/8 \approx 1.178$ for planar (disk) sources. Both endpoints are geometric constants; their derivation is archived with reproduction packages (Fosmark, 2026).

This paper tests that claim completely and honestly, including the failure of one of its registered predictions and the pre-registered repair that followed.

2 The predicted law

The acceleration law tested here is

$$g_{\text{SFT}}^2(r) = g_{\text{bar}}^2(r) + \lambda_{\text{eff}}(r) a_S g_{\text{bar}}(r), \quad (1)$$

equivalent to the interpolating family $\nu(y) = \sqrt{1 + \lambda/y}$. The scale is

$$a_S = \frac{cH_0}{2\pi} = 1.0475 \times 10^{-10} \text{ m s}^{-2}, \quad (2)$$

using $H_0 = 67.74 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck Collaboration, 2020); H_0 is the only imported cosmological quantity and is not adjusted.

The coefficient $\lambda_{\text{eff}}(r)$ interpolates between two geometric endpoints derived from the projection of source maintenance geometry onto the circular-velocity observable:

$$\lambda_{\text{sph}} = \frac{4}{3} \quad (\text{isotropic/bulge sources}), \quad (3)$$

$$\lambda_{\text{disk}} = \frac{3\pi}{8} \approx 1.178 \quad (\text{planar/disk sources}). \quad (4)$$

The disk endpoint incorporates an admissibility factor $\kappa = 3/4$ derived from the binary structure of the theory’s maintenance constraint in 3+1 dimensions. No quantity in Eq. (1) is fitted anywhere in this paper.

3 Why a sealed retest

An earlier analysis in this program profiled a per-galaxy best-fit λ against SPARC rotation curves and reported a morphology–coefficient correlation (Spearman $\rho = -0.27$, $p = 2.8 \times 10^{-4}$) consistent in sign with the geometric band. Our internal re-audit, before submission, found three problems: (i) the correlation is carried by the dwarf-irregular subsample and vanishes within quality-controlled spirals; (ii) only 6.3% of fitted λ values fall inside the predicted band; (iii) the per-galaxy λ absorbs mass-to-light, distance, and functional-form error rather than measuring source geometry.

The fitted- λ observable was therefore retired. The original outputs are preserved in the reproduction package as the motivation for the forward protocol below; the earlier record is superseded, not deleted. We consider the transparent documentation of this self-correction to be part of the scientific content of this work.

4 Protocol

Data partitioning. Inputs were separated before analysis into *source-only* columns (radius; gas, disk, and bulge velocity components; surface-brightness profiles), *target-locked* columns (V_{obs} , eV_{obs} , and all quantities derived from them), and *shared infrastructure* (Hubble type T , quality flag Q). The partition is hash-sealed in the reproduction package.

Model construction. Predictions use Eq. (1) with $\lambda_{\text{eff}}(r) = \sum_j w_j(r) p_j(r) \lambda_j$, where j runs over bulge, stellar disk, and gas components. The weight w_j is each component’s non-negative share of the local baryonic support: $S_{\text{disk}} = \Upsilon_d V_{\text{disk}}^2$, $S_{\text{bul}} = \Upsilon_b V_{\text{bul}}^2$, $S_{\text{gas}} = |V_{\text{gas}}|^2$. Mass-to-light ratios are $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$ at $3.6 \mu\text{m}$ (Schombert et al., 2019). Component coefficients: $\lambda_{\text{bul}} = 4/3$; $\lambda_{\text{star}} = 1.1994$ from the disk endpoint with scale height $R_d/h_z = 7.3$ (Kregel et al., 2002); $\lambda_{\text{gas}} = 4/3$.

Participation weighting. The geometry-only model sets all participation factors $p_j = 1$. The participation-weighted model (Sec. 6) sets $p_{\text{star}} = p_{\text{bul}} = 1$ (collisionless, dynamically recurrent) and

$$p_{\text{gas}}(r) = \frac{V_{\text{src}}^2(r)}{V_{\text{src}}^2(r) + 3\sigma_{1D}^2}, \quad (5)$$

with $V_{\text{src}}^2 = \max(V_{\text{bar}}^2, 0)$ and $\sigma_{1D} = 10 \text{ km s}^{-1}$ (Leroy et al., 2008). This measures the ordered-to-total kinetic support fraction of dissipative gas; the factor 3 converts line-of-sight dispersion to total turbulent support under isotropy and is geometric, not tuned.

The physical motivation is that the anomalous acceleration is sourced by *accumulated dynamical coherence*: how long and how consistently the local baryonic mass has maintained ordered gravitational self-interaction. Collisionless stars on recurrent orbits accumulate this coherence fully over gigayears; turbulent, dissipating gas accumulates only the fraction that is coherently ordered at any given time.

Sealing and dual implementation. Two independent computational engines, built from a common frozen specification with no shared code, generated all predictions from source-only columns. The two sets agree to $\leq 5 \times 10^{-7} \text{ km s}^{-1}$ at every one of 3,391 radial points. Predictions were hash-sealed before the target columns were read. Validation metrics, controls, row rules, eleven subgroup cells, parity threshold, and pass/fail criteria were frozen in writing before unsealing. Scoring was dual-implemented with agreement to ten significant figures. All artifacts are archived with SHA-256 hashes (Fosmark, 2026).

5 Results

Scored rows: 3,389 (two excluded by rule). The best globally fitted constant on these rows is $\lambda^* = 1.156$.

Model	RMS (dex)	Notes
Newtonian only	0.265	
$\lambda = 3\pi/8$ (disk)	0.099	best fixed
$\lambda^* = 1.156$ (fitted)	0.099	1 parameter
Geometry-only SFT	0.099	0 parameters
Participation SFT	0.097	0 parameters

Table 1: Full-sample per-point RMS in $\log_{10} V$. The participation-weighted SFT model defeats the best-fit constant with zero free parameters.

5.1 The acceleration scale

The fitted constant carries the scale result: $\lambda^* a_S = 1.211 \times 10^{-10} \text{ m s}^{-2}$, within 0.9% of the empirical $a_0 = (1.20 \pm 0.02) \times 10^{-10} \text{ m s}^{-2}$ (McGaugh et al., 2016). The derived scale $cH_0/2\pi$ quantitatively realizes the long-noted $a_0 \sim cH_0$ coincidence with a specific geometric coefficient. The fitted coefficient sits 1.9% below the geometric band $[3\pi/8, 4/3]$; the band floor is recovered for $H_0 = 66.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, so the offset is degenerate with the Hubble tension and the mass-to-light normalization at its full size.

5.2 Full-sample comparison

Table 1 reports the per-point RMS of $\log_{10} V_{\text{obs}} - \log_{10} V_{\text{pred}}$ for all models and controls.

5.3 Geometry-only model: wins and a falsification

The geometry-only model achieves 0.45% parity with λ^* and wins four of eleven predeclared subgroups: $T \leq 3$, $T 4-7$, $Q = 1$, and bulge-present galaxies (per-galaxy help fractions 0.54–0.62). This establishes under seal that a morphology-dependent coefficient exists in the predicted direction where data quality is highest.

Its registered prediction that gas-dominated dwarfs prefer $\lambda \rightarrow 4/3$ was **falsified** in both predeclared cells: gas-rich dwarfs prefer coefficients at or below λ^* . The failure localizes to the gas-support sector: instantaneous geometric shape misassigns the boost coefficient for matter that has not dynamically settled.

6 The participation-weighted repair

The successor hypothesis was frozen before any renewed target contact: the anomalous boost is sourced by *accumulated coherent dynamical history*, so dissipative gas banks only its ordered support fraction (Eq. 5). The founding document, conventions, expectations, and falsification criteria were chartered and hashed before the model was scored.

Full-sample result: RMS = 0.097 dex, defeating λ^* (0.099 dex) and every fixed control; per-galaxy help versus λ^* is 57.1%.

Subgroup improvement: The dwarf sector flipped: $T 10-11$ help rose from 0.23 to 0.64; low-gas dwarf cell help rose from 0.21 to 0.70. Quality tiers $Q2$ and $Q3$ converted to wins. Bulge-present and early-type wins were retained.

Standing residuals, reported with equal weight: (i) the eleven most gas-dominated dwarfs received a median effective coefficient of 0.84 where the data prefer $\sim 1.0-1.16$; this brackets a participation *floor* not captured by Eq. (5). (ii) The $Q = 1$ cell marginally breaches the pre-registered parity band at 1.026 versus the 1.02 threshold.

Under the fork’s own charter these residuals grade the repair **partial**; its predeclared kill criteria were not triggered. No parameter was adjusted at any point in response to target data.

7 Interpretation

The results support a specific physical reading: the weak-field anomaly’s *scale* is set by cosmological boundary thermodynamics ($a_S = cH_0/2\pi$, confirmed at the percent level), while its *coefficient* is set by how much of the local baryonic mass has accumulated coherent dynamical history. Collisionless stars on recurrent orbits for gigayears bank that history fully; turbulent gas banks only its coherent fraction; and systems dominated by unsettled gas exhibit weaker boosts.

This replaces the MOND picture in which the anomalous acceleration depends only on instantaneous baryonic acceleration magnitude. It replaces the dark-matter picture in which the anomaly requires a new substance. It predicts a specific, testable dependence on dynamical age and coherence state that neither MOND nor dark matter

provides.

8 Registered predictions

The validated law makes source-computable predictions beyond SPARC, registered here before any test:

Redshift evolution. Because dynamical coherence accumulates, the effective coefficient λ_{eff} at fixed baryonic content rises with cosmic time. High-redshift disks, even those that appear photometrically settled in JWST imaging (Ferreira et al., 2022), should show systematically weaker acceleration boosts if they are kinematically hot (Kassin et al., 2012). The prediction tracks kinematic settling, not apparent morphological shape.

Disturbed systems. Recent mergers and strongly interacting pairs should sit below the RAR relative to settled twins of equal baryonic acceleration, because their dynamical coherence has been disrupted.

Recent accretion. Newly accreted gas is boost-discounted even if dynamically cold at the time of observation, because it has not yet accumulated local coherent history.

Existing high-redshift kinematic surveys (KMOS^{3D}-class, ALMA rotation curves) can test prediction (1) directly.

9 Conclusion

A zero-parameter model, derived from the requirement that spacetime geometry records accumulated irreversible maintenance, predicts a weak-field acceleration law that (1) derives the empirical a_0 scale to 0.9% through $cH_0/2\pi$, (2) matches the best fitted constant within 0.45% on geometry alone while winning predeclared high-quality subgroups, (3) was honestly falsified in gas-rich dwarfs under registered criteria, and (4) was repaired by a pre-registered participation operator that defeats the fitted constant on the full sample.

The sequence documented here—registered prediction, sealed falsification, frozen repair, full-sample pass, standing residuals published as physics—is inseparable from the result. Every number reproduces from hashed artifacts, and the theory’s next obligations (the participation floor in extreme dwarfs, and the $\lambda_{\text{eff}}(z)$ redshift prediction)

are stated before the data that will judge them have been touched.

Acknowledgments

The sealed protocol was executed with two independent AI computational engines (Anthropic Claude; OpenAI GPT-5.5) serving as canonical and adversarial-shadow implementations under the author’s refereeing; their convergences, disagreements, and logged deviations are part of the archived record. The SPARC database is due to Lelli et al. (2016); mass-to-light calibration follows Schombert et al. (2019).

References

- Famaey, B., & McGaugh, S. S. 2012, *Living Rev. Relativ.*, 15, 10
- Ferreira, L., et al. 2022, *ApJL*, 938, L2
- Fosmark, S. 2026, Solvency Field Theory archive, Zenodo, <https://zenodo.org/records/20822219>
- Kassin, S. A., et al. 2012, *ApJ*, 758, 106
- Kregel, M., van der Kruit, P. C., & de Grijs, R. 2002, *MNRAS*, 334, 646
- Lelli, F., McGaugh, S. S., & Schombert, J. M. 2016, *AJ*, 152, 157
- Leroy, A. K., et al. 2008, *AJ*, 136, 2782
- McGaugh, S. S., Lelli, F., & Schombert, J. M. 2016, *Phys. Rev. Lett.*, 117, 201101
- Milgrom, M. 1983, *ApJ*, 270, 365
- Planck Collaboration 2020, *A&A*, 641, A6
- Schombert, J., McGaugh, S., & Lelli, F. 2019, *MNRAS*, 483, 1496
- Verlinde, E. 2017, *SciPost Phys.*, 2, 016